Fiscal Year:	FY 2026	**Task Last Updated:**	FY 10/29/2025
PI Name:	Fornace, Albert M.D.		
Project Title:	Space Radiation-Induced Persistent Estrogenic Response and Risk of Breast Cancer Development		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **SR**:Space Radiation		
Human Research Program Risks:	(1) **Cancer**:Risk of Radiation Carcinogenesis		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Zip Code:	20007-2126	**Congressional District:**	1
Comments:	http://www9.georgetown.edu/		
Project Type:	Ground	**Solicitation / Funding Source:**	2018 HERO 80JSC018N0001-Crew Health and Performance (FLAGSHIP, OMNIBUS). Appendix A-Flagship, Appendix B-Omnibus
Start Date:	01/02/2020	**End Date:**	05/01/2026
No. of Post Docs:	1	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JSC
Contact Monitor:	Zawaski, Janice	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 05/01/2026 per V. Lehman/NASA-JSC (Ed., 10/8/25). NOTE: End date changed to 05/01/2025 per S. Mack-Phillips/NASA-JSC (Ed., 5/8/24).		
Key Personnel Changes/Previous PI:	Ed. note (December 2020): The original Principal Investigator (PI) was Kamal Datta, M.D., who is now affiliated with NIH, as of early 2019. PI changed to Albert Fornace, M.D., before grant was awarded. In January 2023, the Co-Investigator (Dr. Michael Johnson) was replaced with Dr. Marc E. Lippman, who is a world-renowned breast cancer expert and Professor of Oncology at the Lombardi Comprehensive Cancer Center at Georgetown University.		
COI Name (Institution):	Brenner, David Ph.D. (Columbia University) Li, Xin Ph.D. (Georgetown University) Suman, Shubhankar Ph.D. (Georgetown University) Lippman, Marc (Georgetown University)		
Grant/Contract No.:	80NSSC19K1649		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Based on epidemiological analysis of radiation-exposed human cohorts, it is projected that space radiation could significantly increase breast cancer incidence and mortality among female astronauts. However, there is a large uncertainty in breast cancer risk estimation due to a lack of in vivo data. The proposed study aims to acquire in vivo quantitative and qualitative data on space radiation-induced breast cancer risk using a mouse model approach. Further, we also aim to test “persistent estrogenic response” (PER) signaling as a target to develop a potential countermeasure against space radiation-induced mammary tumorigenesis. We further hypothesize that targeting GCR-induced PER using an FDA-approved selective estrogen receptor modulator (SERM) will decrease the risk of mammary tumorigenesis. We plan to test our hypothesis by pursuing the following specific aims: Aim 1. Quantitatively compare effects of simulated GCR dose rates on mammary tumor incidence and grade in the ApcMin/+ tumor model. Aim 2. Characterize GCR dose-rate effects on PER in relation to mammary tumorigenesis. Aim 3. Determine roles of SERM in countering space radiation-induced PER and mammary tumorigenesis. Aim 4. Risk assessment (modeling) of mammary tumorigenesis after space radiation exposure. Key deliverables for this project are: 1) Develop a mathematical model for GCR dose rate-based breast cancer risk estimation, 2) Identify early markers of GCR-induced mammary tumorigenesis, and 3) Test potential mechanism-based FDA-approved countermeasure agents.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Breast cancer is the second most common cancer among women in the US. According to current breast cancer incidence data, approximately 12.9 percent of women in the United States will be diagnosed with breast cancer at some point in their lives. The annual rate of new cases and mortality from breast cancer in the United States is 128.5 per 100,000 and 20.1 per 100,000 women, respectively. Considering the high frequency of breast cancer in the American population, even a small increase by space radiation could have a major impact on risk estimates and planning of future space missions. In addition to risk estimation, studies on persistent estrogenic response after space radiation and its role in driving mammary tumorigenesis may provide insight into signaling events affecting carcinogenesis. The significance and deliverables of this project are to improve the estimates of breast cancer risk in women astronauts and to identify and test plausible targets for the development of mitigation strategies.
Task Progress:	1. Female mice exposed to 33-ion full-spectrum simulated galactic cosmic radiation (GCRsim) developed mammary (breast) tumors at a much higher rate than those exposed to conventional gamma radiation. Both acute (short-term) and chronic (prolonged) GCR exposures produced similar cancer rates, showing that simply lowering the exposure rate does not reduce biological harm. These results indicate that space radiation causes a strong, dose-dependent cancer risk that is largely independent of dose rate, underscoring the importance of developing biological countermeasures for astronaut protection. 2. GCRsim-exposed mice showed elevated estrogen levels in their blood and increased activity of estrogen receptor alpha (ERα) and estrogen-related receptor alpha (ERRα) in mammary tissue. These receptors regulate genes that control cell growth, including Cyclin D1 and secreted phosphoprotein 1 (SPP1 or osteopontin), molecules known to promote breast tumor development in women. The coordinated upregulation of ERα, ERRα, and SPP1 suggests that space radiation enhances hormone-related signaling pathways, which may accelerate radiation-induced breast cancer. 3. Comprehensive proteomic analysis of blood samples from GCRsim-exposed mice identified hundreds of altered proteins, many linked to inflammation, immune activation, and cancer progression. Several of these were classified as senescence-associated secretory phenotype (SASP) factors, which are released by stressed or aging cells and can promote chronic inflammation. At the same time, levels of key antioxidant enzymes (such as SOD1, GPX2, and catalase) were reduced, indicating oxidative stress plays a central role in radiation-induced tumor development. Together, these findings highlight a dual process of persistent inflammation and impaired antioxidant defense after space radiation exposure. 4. Elevated serum levels of SPP1 (osteopontin) and several other radiation-responsive proteins were strongly associated with increased cancer incidence. These molecules could serve as early, non-invasive biomarkers to assess individual susceptibility to space radiation–induced cancer. Such markers may help develop screening tools to monitor astronaut health during and after deep space missions and to guide personalized countermeasure strategies. 5. To identify potential protective treatments, researchers developed a novel ex vivo mouse mammary organ culture (MMOC) model that mimics early radiation-induced precancerous changes. Using this model, several cancer-preventive compounds, including the senolytic drug fisetin and hormone-targeting agents (tamoxifen and anastrozole), significantly reduced radiation-induced cellular senescence, inflammation, and abnormal tissue growth. The MMOC system proved to be a powerful and reproducible platform for studying early tumor development and for screening pharmacological countermeasures against space radiation–associated breast cancer risk. 6. Collectively, these findings provide compelling evidence that space radiation poses a serious risk for hormone- and inflammation-driven cancers through mechanisms involving cellular senescence, oxidative stress, and dysregulated hormonal signaling. The identification of serum biomarkers and development of testable prevention models represent important steps toward risk surveillance and mitigation strategies aimed at safeguarding astronaut health during long-duration missions beyond low Earth orbit.
Bibliography Type:	Description: (Last Updated: 06/30/2026)
Abstracts for Journals and Proceedings	Kumar K, Moon BH, Kumar S, Angdisen J, Kallakury BVS, Fornace AJ Jr, Suman S. "Senolytic agent ABT-263 mitigates low- and high-let radiation-induced gastrointestinal tumor development." 2025 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025
Abstracts for Journals and Proceedings	Kumar K, Angdisen J, Ma J, Datta K, Kallakury BVS, Fornace AJ Jr, Suman S. "High risk of mammary cancer from space radiation exposure: Evidence from Apcmin/+ mouse studies." 2025 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, January 28-31, 2025. Abstracts. 2025 NASA Human Research Program Investigators’ Workshop, Galveston, Texas, January 28-31, 2025. , Jan-2025

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Abstracts for Journals and Proceedings	Suman S. "From Earth to deep space: Insight into animal model on radiation health risk (invited talk-ONLINE)." School-cum-Workshop on Biological Effects of Ionizing Radiation, Inter-University Accelerator Centre (IUAC), New Delhi, Delhi, India, September 24-26, 2025. Abstracts. School-cum-Workshop on Biological Effects of Ionizing Radiation, Inter-University Accelerator Centre (IUAC), New Delhi, Delhi, India, September 24-26, 2025. , Sep-2025
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